NoSQL and MongoDB

Structured vs. Semi-Structured vs. Unstructured Data

- **Structured Data**

Data that is **organized in a fixed format** (rows & columns).
It follows a **strict schema**.
Characteristics:
Stored in tables
Easy to search with SQL
Clearly defined fields
Examples:
Student database table
Bank transactions
Employee records
Used In:
Relational databases (MySQL, PostgreSQL)
Banking systems

- **Semi-Structured Data**

Data that does **not follow a strict table structure**,
but still has some organization (tags, keys).

Flexible schema.

Characteristics:

Self-describing

Uses tags or key-value pairs

Not stored in traditional tables

Examples:

JSON

XML

NoSQL documents

Used In:

NoSQL databases (MongoDB)

APIs

Web applications

```json
{
  "name": "Ahmed",
  "age": 20,
  "courses": ["DB", "AI"]
}
```

Notice: Another student may have different fields.

- **Unstructured Data**

Data with **no predefined format or structure**.
Cannot be stored easily in tables.

Characteristics:

Large volume
Difficult to search directly
Needs special tools (AI, ML)

Examples:

Images
Videos
Audio files
Emails
Social media posts
PDF documents

Used In:

Social media platforms
Multimedia systems
Big data analytics